

Review

Epigenetics and Exercise

Sean L. McGee^{1,*} and Mark Hargreaves^{2,*}

Epigenetics can be defined as ‘the structural adaptation of chromosomal regions so as to register, signal, or perpetuate altered activity states.’ Increased transcription of key regulatory, metabolic, and myogenic genes is an early response to exercise and is important in mediating subsequent adaptations in skeletal muscle. DNA hypomethylation and histone hyperacetylation are emerging as important crucial events for increased transcription. The complex interactions between multiple epigenetic modifications and their regulation by metabolic changes and signaling events during exercise, with implications for enhanced understanding of the acute and chronic adaptations to exercise, are questions for further investigation.

Background

The Oxford dictionary defines epigenetics as ‘the study of changes in organisms caused by the modification of gene expression rather than alteration of the genetic code itself.’ Over the years in biology it has had several meanings, from how genotypes give rise to phenotypes during development, to heritable changes in gene function not attributable to changes in gene sequence [1]. A unifying definition that sought to reconcile contemporary usages with the requirement of heritability has been proposed [1]: ‘the structural adaptation of chromosomal regions so as to register, signal, or perpetuate altered activity states.’ The two most studied epigenetic events are DNA methylation and post-translational modification of histones, the major protein component of chromatin responsible for organizing DNA within nucleosomes (Figure 1). Other potential epigenetic modifications are those mediated by noncoding RNAs, notably microRNAs. These microRNAs can alter gene expression via post-transcriptional modulation and may also influence translational events. Identification of circulating microRNAs has raised the possibility that they are involved in cell–cell and tissue–tissue communication.

Exercise has profound effects on all physiological systems within the body [2] and regular physical activity/exercise training results in enhanced functional capacity and improved health and wellbeing. Much attention has focused on skeletal muscle responses and adaptations to acute and chronic exercise stimuli, since these adaptations contribute significantly to improved athletic performance and health. A complex array of molecular mechanisms, involving various signaling pathways and downstream mediators, has been identified in skeletal muscle [2,3]. In addition, contracting skeletal muscle releases a number of biologically active proteins, nucleic acids, and metabolites (‘myokines’) that may mediate some of the systemic effects of exercise [4]. Over the years, a paradigm has emerged that highlights the importance of transient increases in the mRNA levels of various metabolic, myogenic, and regulatory genes in response to a single bout of exercise, which when repeated on a regular basis as occurs during exercise training, leads to increased levels of key proteins that mediate many of the functional benefits of such training [3,5,6]. Experimental evidence in support of this paradigm has been reported [7–9]. Of course, enhanced translational capacity and proteome abundance are also key elements of the increased skeletal muscle metabolic function and mass following exercise training [10]. Nevertheless, much attention has focused on the early transcriptional events and their regulation in skeletal muscle in response to a single exercise bout and the potential contribution of epigenetic mechanisms. This

Highlights

Alterations in epigenetic marks, including DNA methylation and histone modifications, are linked to the skeletal muscle transcriptional responses to exercise that ultimately mediate the exercise adaptations.

Signaling pathways activated by exercise control the activity of epigenetic modifying enzymes, providing specificity to the exercise-induced transcriptional response.

Flux through metabolic pathways can impact on epigenetics directly by modulating the availability of the metabolic intermediates that are required for epigenetics marks, such as acetyl-CoA for acetylation and S-adenosyl-L-methionine for methylation. Furthermore, some metabolites, such as lactate, can directly regulate the activity of epigenetic modifying enzymes.

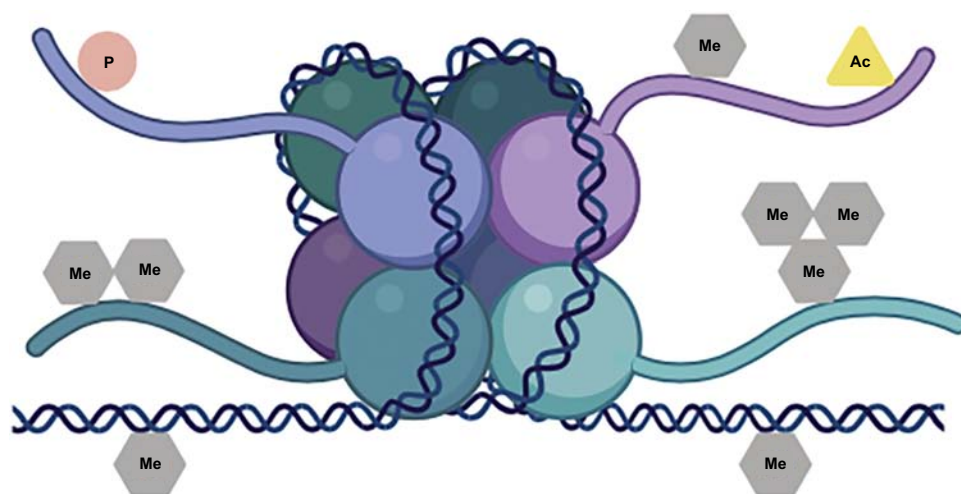
Emerging evidence suggests that epigenetic modifications can mediate the inter-generational transmission of exercise and diet effects on physiology.

¹Metabolic Research Unit, School of Medicine and Centre for Molecular and Medical Research, Deakin University, Geelong Waurn Ponds, VIC 3216, Australia

²Department of Physiology, The University of Melbourne, VIC 3010, Australia

*Correspondence: sean.mcgee@deakin.edu.au (S.L. McGee) and m.hargreaves@unimelb.edu.au (M. Hargreaves).





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Figure 1. Major Types of Epigenetic Regulation. DNA methylation and histone (blue, green, and purple circles) post-translational modifications are capable of regulating gene transcription. Abbreviations: Me, methylation (mono-, di-, and tri-methylated lysine residues; mono- and di-methylated arginine residues); P, phosphorylation; Ac, acetylation. Figure prepared using BioRender.

review provides a brief overview of exercise effects on DNA methylation and histone modifications, predominantly from studies in humans. It will not address the contribution of microRNAs and readers are referred to a recent review on this topic [11].

DNA Methylation

DNA methylation involves the addition of a methyl group to the 5-carbon position of cytosine bases through the action of a family of DNA methyltransferases (DNMT). The cytosine bases most susceptible to methylation are often found in the cytosine-phosphate-guanine (CpG) dinucleotide sequences of DNA, referred to as CpG islands. Studies from both plants and animals have shown that the major biological consequence of DNA methylation is gene silencing [12]. It is also likely to be involved in the etiology of human disease. As an example, cytosine hypermethylation of peroxisome proliferator-activated receptor γ coactivator-1 α (*PGC-1 α*) promoter region is associated with reduced skeletal muscle mitochondrial content in patients with type 2 diabetes [13]. DNA methylation can directly prevent the binding of basal transcriptional machinery or of most mammalian transcription factors that require interactions with cytosine. These transcription factors, such as Sp1, have GC-rich binding sites and CpGs in their DNA recognition elements [12]. Methylated CpGs can also influence nucleosome stability or positioning and core histone access, thereby modifying access of transcription factors to promoter regions.

Given the increase in skeletal muscle mRNA levels of selected genes following a single bout of exercise, it is possible that changes in DNA methylation play a role in exercise-induced gene expression. Indeed, acute exercise in human subjects resulted in a reduction in global DNA methylation [14]. Further analysis of gene-specific DNA methylation indicated variable responses; however, for key metabolic and regulatory genes, such as *PGC-1 α* , peroxisome proliferator-activated receptor δ (*PPAR- δ*), mitochondrial transcription factor A (*TFAM*), and myocyte enhancer factor 2 (*MEF2*), exercise-induced DNA hypomethylation was generally associated with increased gene expression, with effects being greater at higher exercise intensities [14]. Consistent with this

finding was the observation that acute exercise resulted in hypomethylation of the skeletal muscle *PGC-1 α* promoter and altered nucleosome structure around this hypomethylated site, which correlated with the fold increase in *PGC-1 α* gene expression and the postexercise decrease in intramuscular lipid content [15]. The underlying mechanisms mediating exercise-induced changes in DNA methylation remain to be fully elucidated. *Ex vivo* contraction in isolated rat soleus resulted in DNA hypomethylation and increased gene expression, suggesting intramuscular factors may be primary [14]. Caffeine produced similar responses in L6 myotubes, implicating calcium in the signaling between exercise and DNA methylation [14]. A single exercise bout produced dynamic changes in DNA methylation, with 130 hypomethylated and 110 hypermethylated regions in human adipose tissue [16]. There was only a modest association between DNA methylation and gene expression, suggesting that gene expression is only partly regulated by DNA methylation. Furthermore, despite acute exercise-induced gene expression changes being smaller in magnitude following 6 weeks of endurance training, the number of differentially methylated regions was more pronounced [16]. Again, there was not a clear association between the changes in DNA methylation and gene expression following training, which supports the idea that gene expression is only partly regulated by DNA methylation. In contrast, in skeletal muscle endurance, exercise training does appear to be associated with a relative DNA hypomethylation of key metabolic and regulatory genes, such as NADH:ubiquinone oxidoreductase subunit C2 (*NDUFC2*) and *MEF2A* [17,18]. An interesting observation after a second period of resistance training, following an initial resistance training program and subsequent detraining/unloading, was an increased frequency of hypomethylation with reloading (~18 800 CpG sites) compared with the initial loading (~9100 CpG sites), suggesting the potential of an 'epigenetic memory' of hypertrophy in human skeletal muscle [19]. The significance of this remains to be determined, although the concept of an exercise 'epigenetic memory' is not supported by well-controlled experiments examining the transcriptional response to endurance exercise [20]. Finally, a cross-sectional comparison of elite athletes and sedentary control subjects demonstrated polymorphisms in a number of enzymes involved in DNA methylation and synthesis in the elite athletes that contributed to a relative DNA hypomethylation and perhaps a greater potential to increase muscle mass [21]. Collectively, the results suggest that exercise generally results in DNA hypomethylation of key genes in skeletal muscle, and this represents an early response mediating skeletal muscle adaptations to exercise.

Histone Modifications

Although the skeletal muscle transcriptional responses to different exercise modalities has been well defined over the past decade, our understanding of the epigenetic mechanisms involved in regulating these responses is still rudimentary (Box 1). Exercise increases skeletal muscle histone 3 (H3) serine phosphorylation in both untrained and trained subjects [22]. Our own early work in this area identified global increases in H3 lysine (K36) acetylation in human skeletal muscle following a single 60 min bout of cycling at ~70% $\text{VO}_{2\text{peak}}$ [23]. However, these studies did not examine the chromatin regions at which these histone modifications occurred. Indeed, there are only a handful of studies to date that have examined exercise-induced histone modifications at specific gene loci by chromatin immunoprecipitation (ChIP). Swimming exercise in rats increased H3 K9/14 acetylation, a histone mark associated with transcriptional activation, at the glucose transporter type 4 (*Glut4*) promoter in triceps muscle [24]. Similar data has been observed for the nuclear respiratory factor 1 (*Nrf-1*) and *Mef2A* promoter regions [25]. A recent study has examined histone acetylation and other epigenetic regulatory mechanisms across different regions of the *Pgc-1 α* gene in response to treadmill running in rats. Interestingly, histone acetylation and RNA polymerase II (pol II) abundance across different regions of the *Pgc-1 α* gene correlated poorly with the exercise-induced increase in *Pgc-1 α* gene expression in both plantaris and soleus skeletal muscles, which are comprised of predominantly fast and slow fibers, respectively. However,

Box 1. DNA, Histones, and Nucleosome.

DNA is wrapped around an octamer of histones 2A, 2B, 3, and 4, which is referred to as a nucleosome [78]. Repeating nucleosomes together form chromatin, which has a complex three-dimensional structure that allows vast lengths of DNA to be efficiently packaged into the nucleus [78]. The structure of chromatin also plays a critical role in transcriptional regulation. In the transcriptionally repressed state, chromatin is densely compacted such that the transcriptional machinery, including polymerase II (Pol II), cannot gain access to promoter and gene regions to initiate transcription [79]. Specific post-translational modifications of histone proteins can decompact chromatin, which also requires histone exclusion or exchange from the nucleosome by chromatin remodeling enzymes and histone chaperones [80]. This state is generally permissive for transcriptional activation by exposing promoter and gene regions to the transcriptional machinery that initiates gene expression [79]. Histone modifications that activate transcription include: phosphorylation, acetylation, and methylation, as well as other acylation marks, which are emerging as novel histone post-translational modifications [81]. Lysine residues play an important role in the regulation of chromatin structure and function, as in the unmodified state the lysine side chain carries a positive charge that is the basis for histone–DNA and nucleosome–nucleosome interactions. Histone modifications, particularly acetylation by histone acetyltransferases (HATs), neutralizes the charge on the lysine side chain and is sufficient to break these electrostatic interactions in what is the first step of histone exclusion and transcriptional initiation [80]. However, not all histone modifications confer transcriptional activation and some instead favor transcriptional repression, including methylation of certain histone regions [82]. An overview of characterized histone modifications and their generalized effect on transcription has been reviewed recently [83]. These histone modifications are regulated by a network of histone modifying enzymes, the activity of which is regulated by intracellular signaling mechanisms that link extracellular stimuli with appropriate transcriptional responses. Conversely, transcriptional deactivation is controlled by transcriptional repressors, including histone deacetylases (HDACs), which promote the remodeling of chromatin back into its compacted state by removing histone modifications that initiate transcription. These modifications are transient, with the half-life of acetylated histones being ~3 min and phosphorylated histones ~20 min [84].

the actual abundance of H3 across the *Pgc-1 α* gene was the best predictor of *Pgc-1 α* gene expression, suggesting that histone exchange patterns could play an important role in regulating exercise-induced transcriptional responses. Furthermore, an increase in H3 K4 trimethylation during exercise at the alternative upstream *Pgc-1 α* promoter was associated with expression of the alternative *Pgc-1 α* transcript [26]. This same histone modification was not observed at the canonical promoter and this study remains the only one to examine a specific histone methylation mark at a promoter level following exercise. Intriguingly, a number of studies have examined histone modifications in various brain regions in response to exercise. Forced swimming induces H3 K14 acetylation and serine 10 phosphorylation in spatially distinct regions of the dentate gyrus in a time-dependent manner [27]. Furthermore, voluntary exercise in rats increases H3 acetylation at the brain-derived neurotrophic factor (*Bdnf*) promoter in the hippocampus [28]. These studies were among the first to associate changes in histone modifications with alterations in exercise-induced gene expression. It is possible that the endocrine functions of skeletal muscle during exercise that link muscle contraction with adaptive responses to exercise in tissues, such as the liver and adipose tissue, could also mediate an exercise transcriptional response in the brain.

Although there is a paucity of information describing the specific histone modifications regulating gene expression responses to exercise, the signaling mechanisms involved in the adaptive response to exercise in skeletal muscle are better defined. Pathways, including the AMP-activated protein kinase (AMPK), mitogen activated protein kinases (MAPK), protein kinase A (PKA), protein kinase C (PKC), and the calcium/calmodulin protein kinase II (CaMKII), are all important for phosphorylation-dependent signaling in skeletal muscle during exercise [29]. A number of studies have linked these signaling pathways with specific histone modifications. A critical stimulus driving exercise adaptations is reduced cellular energy balance, as muscle contraction markedly increases ATP requirements. The heterotrimeric AMPK complex is an important sensor of low cellular energy balance, and complexes containing the α_2 catalytic subunit are activated by endurance-based exercise of moderate intensity [30]. Activation of AMPK induces transcriptional responses that provide protection against future energy perturbations, including mitochondrial biogenesis and increased transport and handling capacity of glucose and fatty acids [31–33].

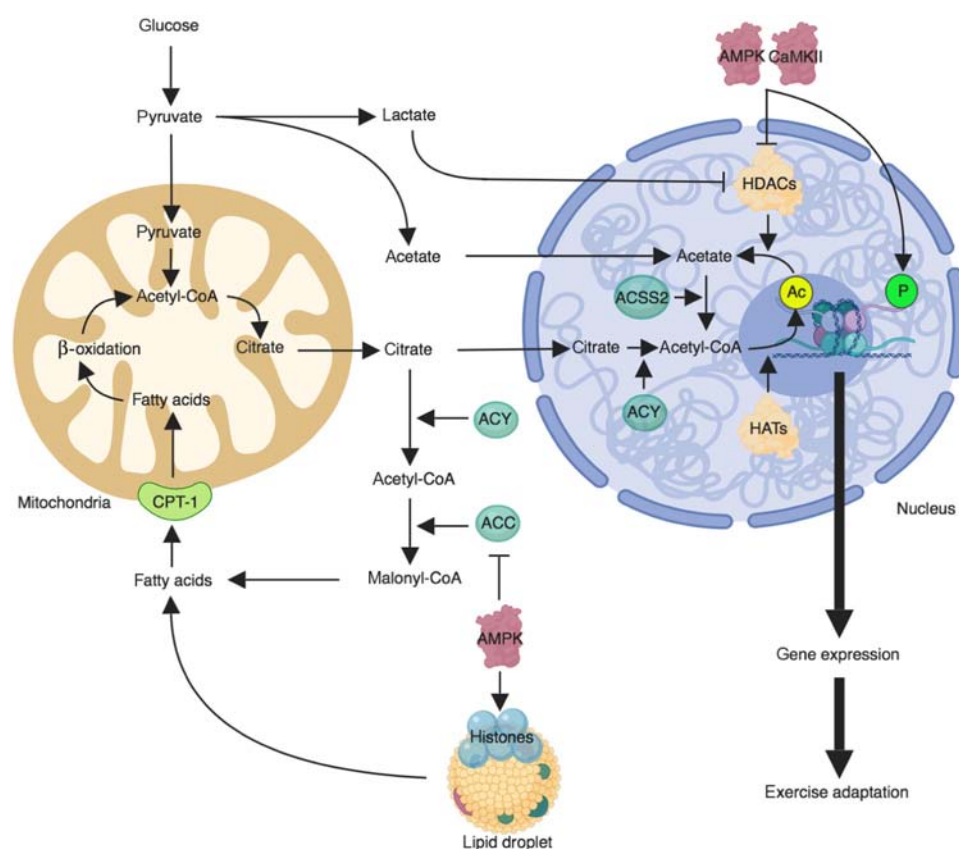
An extensive compendium of AMPK substrates now exists and a number of important regulators of histone modifications have been identified as bona fide AMPK substrates [34,35]. Although much of this work has not been verified in skeletal muscle or in the exercise adaptive response, these mechanisms give insights into potentially important pathways by which epigenetics and transcription are controlled by exercise. Our own work has established the class IIa histone deacetylases (HDACs) as AMPK substrates, with phosphorylation resulting in their nuclear export and derepression of the MEF2 transcription factor [36,37]. Class IIa HDAC phosphorylation following AMPK activation results in acetylation of H3 at K9 and 14 [36], modifications that are associated with transcriptional initiation. The class IIa HDACs do not possess catalytic activity against acetyl-lysine due to a critical amino acid substitution in the otherwise homologous HDAC active site [38], but instead recruit a corepressor complex that includes HDAC3 [39]. Indeed, genetic and pharmacological disruption of the class IIa HDAC corepressor complex induces exercise-like transcriptional responses in skeletal muscle [40]. Regulation of class IIa HDAC phosphorylation during exercise also highlights the redundancy in the signaling mechanisms controlling important exercise adaptive responses. In AMPK α_2 dominant negative mice, class IIa HDAC phosphorylation was maintained during exercise by a compensatory increase in protein kinase D (PKD) activation [41], an alternative class IIa HDAC kinase [42]. Although PKD is not normally activated by exercise [23], expression of constitutively active PKD in myogenic cells increases the expression of exercise-responsive genes in a class IIa HDAC phosphorylation-dependent manner, including *PGC-1 α* , *PPAR α* , and *GLUT4* [41]. Furthermore, CaMKII is also a class IIa HDAC kinase [43] that has been linked to gene expression responses to exercise. Administration of a moderately specific CaMKII inhibitor to rats prior to an acute bout of exercise is sufficient to prevent histone acetylation at the *GLUT4* promoter around its MEF2 binding site [24]. In addition to kinase control of HDAC function, a number of metabolites produced during exercise, including lactate [44] by skeletal muscle and β -hydroxybutyrate [45] by the liver, can act as direct inhibitors of HDACs at high physiological concentrations. For example, it has been reported that the increase in BDNF expression in the brain following exercise is due to the inhibitory action of β -hydroxybutyrate on HDAC2 and 3 at the *Bdnf* promoter [46]. The contribution of lactate to epigenetic control during exercise remains to be effectively quantified.

Interestingly, both AMPK and CaMKII also phosphorylate H3 directly [34,47]. It is thought that phosphorylation of this site is required before H3 can be acetylated [48], highlighting the stepwise control of chromatin decompaction and histone exclusion mechanisms that are required to initiate transcription. It also highlights that signaling pathways mediating exercise adaptations, and transcriptional control more broadly, typically regulate multiple epigenetic mechanisms. Redundancy could also explain many of the findings from exercise studies in histone acetyltransferase (HAT) loss of function in animal models, which collectively have failed to definitively identify HATs that mediate chromatin remodeling during exercise. For example, knockout of p300 has no effect on the skeletal muscle adaptations to exercise training [49], despite p300 being one of the most important acetyltransferases mediating histone acetylation [50]. Indeed, there appears to be considerable overlap in substrate specificity between the five major acetyltransferase families [51]. Similar findings have been observed for the transcriptional coactivator PGC-1 α , which was thought to be an essential contributor to the transcriptional response to exercise through its ability to recruit HATs to specific transcription factors [52,53]. However, skeletal muscle-specific *Pgc-1 α* knockout mice have normal skeletal muscle transcriptional responses to acute exercise [54,55]. These studies highlight the importance of the adaptive response to exercise and the apparent redundancy designed to protect this response. Indeed, organismal survival throughout evolution was highly dependent on the ability to adapt to the energetics demands imparted by the need for safety, food, and shelter. So fundamental for survival is this trait, it

appears that molecular redundancy has evolved to protect it. Given this redundancy, new approaches beyond simple genetic manipulation will be required to tease out the signaling and epigenetic mechanisms controlling the exercise adaptive response.

Metabolism and Epigenetic Modifications

Investigations over the past decade are beginning to detail interactions between metabolic pathway flux and the regulation of epigenetic modifications (Figure 2). For example, acetylation and methylation reactions require metabolite intermediates from specific metabolic pathways in the form of acetyl-CoA and S-adenosyl-L-methionine (SAME), respectively. It appears that a variety of substrates can provide the acetyl-CoA required for histone acetylation in different cell types and under different energetic contexts. Initial studies in transformed cells suggested that glucose was the predominant substrate providing the nucleocytosolic acetyl-CoA required for histone acetylation [56]. Although glucose-derived acetyl-CoA is produced in the mitochondria, mitochondrial citrate is exported to the cytosol via the citrate transporter before being converted



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Figure 2. Established and Potential Interactions between Intermediary Metabolism and Histone Modifications.

Metabolic pathways that produce acetyl-CoA are linked to histone acetylation (Ac) through the actions of histone acetyltransferases (HATs), while histone deacetylases (HDACs) oppose this process. Key enzymes that produce acetyl-CoA from citrate and acetate are ATP-citrate lyase (ACLY) and acetyl-CoA synthetase 2 (ACSS2). Exercise-responsive kinases, including the AMP-activated protein kinase (AMPK) and the calcium/calmodulin-dependent protein kinase II (CaMKII), phosphorylate (P) histones. Regulation of histones associated with lipid droplets might also play a role in the supply of fatty acids. Abbreviations: ACC, acetyl-CoA carboxylase; CPT-1, carnitine palmitoyl-transferase 1. Figure prepared using BioRender.

back to acetyl-CoA by ATP-citrate lyase (ACLY) [56]. A recent study has also suggested that acetyl-CoA synthetase (ACSS2), which converts acetate to acetyl-CoA, is also important for histone acetylation [57]. *De novo* acetate production was thought to be negligible in mammalian cells until a recent study identified a new pathway of acetate synthesis directly from pyruvate, through two distinct mechanisms that involve either reactive oxygen species or keto acid dehydrogenases [58], further linking glucose utilization and histone acetylation. Studies in nontransformed cells suggest that the majority of acetyl-CoA required for histone acetylation is derived from fatty acid oxidation [59]. Intriguingly, using stable isotope tracking of fatty acid-derived carbon, fatty acid-mediated histone acetylation was linked to specific transcriptional reprogramming of lipid metabolism [59]. It is noteworthy that AMPK also regulates histone acetylation through the control of metabolic pathway flux. AMPK is a well-established inhibitor of the acetyl-CoA carboxylase (ACC) enzyme that converts acetyl-CoA to malonyl-CoA, the first step in fatty acid synthesis. Malonyl-CoA is an allosteric inhibitor of the carnitine palmitoyl-transferase 1 (CPT-1) transporter of fatty acids into the mitochondria. While phosphorylation-dependent regulation of ACC is the major mechanism by which AMPK increases β -oxidation of fatty acids, including during exercise [60], ACC inhibition also increases the amount of acetyl-CoA available for histone acetylation [61,62]. This highlights the coordinated control of multiple epigenetic regulatory mechanisms by signaling mediators of the epigenetic responses.

It has also been speculated that the interaction between metabolism and histone acetylation could be bidirectional. Based on the number of known histone acetylation sites, chromatin can consume up to 3 mM of acetyl-CoA, while the cellular concentration of acetyl-CoA is maintained around 20 μ M [63]. As deacetylation reactions produce acetate, this suggests that histones are a large reservoir of this metabolite, which can be used to regenerate nucleocytosolic acetyl-CoA for metabolic or signaling purposes. Acetate also has important roles in the regulation of cellular pH [64]. Histones have also been identified in the proteome of lipid droplets in a variety of cell types [65–68]. While these initial findings focused on the ability of lipid droplets to control the supply of histones throughout development [66] and as a store that can release histones that act as antimicrobial agents [68], it is also possible that control of histone function is important for regulation of fatty acid metabolism. The interaction between metabolism and histone acetylation raises a number of important questions regarding exercise-mediated epigenetic regulation that remain to be resolved. For example, how is histone acetylation increased at exercise-responsive gene regions during exercise when there is high energetic demand? What substrates supply this acetyl-CoA? Does histone acetylation during exercise require acetyl-CoA production, or is it simply dependent on acetate redistribution within distinct chromatin regions? Does the histone acetate reservoir play a role in skeletal muscle metabolism or pH regulation during exercise? Is the regulation of histones at lipid droplets important for fatty acid metabolism during exercise? These are just a few of the questions that will shape research within the field in the coming years. Similarly, the DNA demethylation events that have been characterized during exercise could also have important metabolic implications. Demethylation reactions produce formaldehyde from the liberated methyl group, which through its detoxification produces reducing equivalents in the form of NADH [63], which could be coupled to mitochondrial ATP production. Chromatin can consume up to 11 mM of S-adenosylmethionine (SAMe) [63] and could therefore provide a significant energy source during exercise. It is clear that the field is only just scratching the surface with regard to the important interactions between epigenetics and metabolism and how they influence exercise adaptive responses.

Intergenerational Transmission

A key question is whether the effects of exercise in parents can be transmitted to offspring and, if so, whether this is mediated by epigenetic modifications. It is well recognized that the parental, intrauterine, and postnatal environments can have profound effects on the health outcomes of

offspring and this has led to testing of the ‘developmental origins of health and disease hypothesis.’ Perinatal, maternal exercise has been shown to improve the metabolic health of offspring [69–71] and to prevent the deleterious effects of a maternal high-fat diet on the metabolic health of offspring [72]. Paternal exercise has also been shown to improve glucose metabolism in offspring, related to improved sperm motility and alterations in multiple classes of small RNAs in sperm [73]. Preconception exercise in obese fathers has been shown to normalize the sperm microRNA profile and improve the metabolic health of offspring [74]. Other studies have observed exercise-induced changes in sperm microRNA expression and methylation in relation to genes associated with neurological function [75,76]. Finally, exercise prevented hypermethylation of the *Pgc-1 α* promoter in offspring of mothers on a high-fat diet, which resulted in enhanced *Pgc-1 α* and its target gene expression and improved metabolic function in offspring [77]. Collectively, these results implicate epigenetic modifications as mediators of intergenerational transmission of exercise and diet effects. Given the increasing prevalence of obesity and metabolic dysfunction and the impact that they have on the development of a number of chronic diseases, this is an important area for future investigation. Defining the specific mechanisms mediating the benefits of maternal and paternal exercise on offspring will be an important step in developing optimized exercise interventions to improve the health and wellbeing of potential parents and to mitigate transmission of such characteristics to their offspring. Such endeavors will be vital to attenuate the increasing prevalence of chronic disease in modern society.

Concluding Remarks

Increased transcription of key regulatory, metabolic, and myogenic genes is an early response to exercise and is important in mediating subsequent adaptations in skeletal muscle. DNA hypomethylation and histone hyperacetylation are emerging as important crucial events for increased transcription. There are likely to be complex interactions between these epigenetic modifications and this is an area for future research, together with investigation of the metabolic changes and signaling events during exercise that may influence such modifications. Enhanced understanding of these interactions will inform optimization of exercise interventions and the development of novel therapeutic strategies to manage metabolic disease (see Outstanding Questions).

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Outstanding Questions

What are the specific epigenetic marks that activate transcription of genes during exercise, and what are the epigenetic modifying enzymes involved?

How is histone acetylation increased at exercise-responsive gene regions during exercise when there is high energetic demand? What substrates supply this acetyl-CoA? Does histone acetylation during exercise require acetyl-CoA production, or is it simply dependent on acetate redistribution within distinct chromatin regions?

Does the histone acetate reservoir play a role in skeletal muscle metabolism or pH regulation during exercise?

Is the regulation of histones at lipid droplets important for fatty acid metabolism during exercise?

What are the specific epigenetic mechanisms involved in the intergenerational transmission of exercise adaptations? Are these adaptations persistent, or are they lost with inactivity?

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